

Grid-Supporting Equipment Supply Chains as a Constraint on Power System Expansion

Transformer can become a bottleneck before planned grid expansion is realized

Boyu Yao^{1,3}, Yury Dvorkin^{1,2,3}

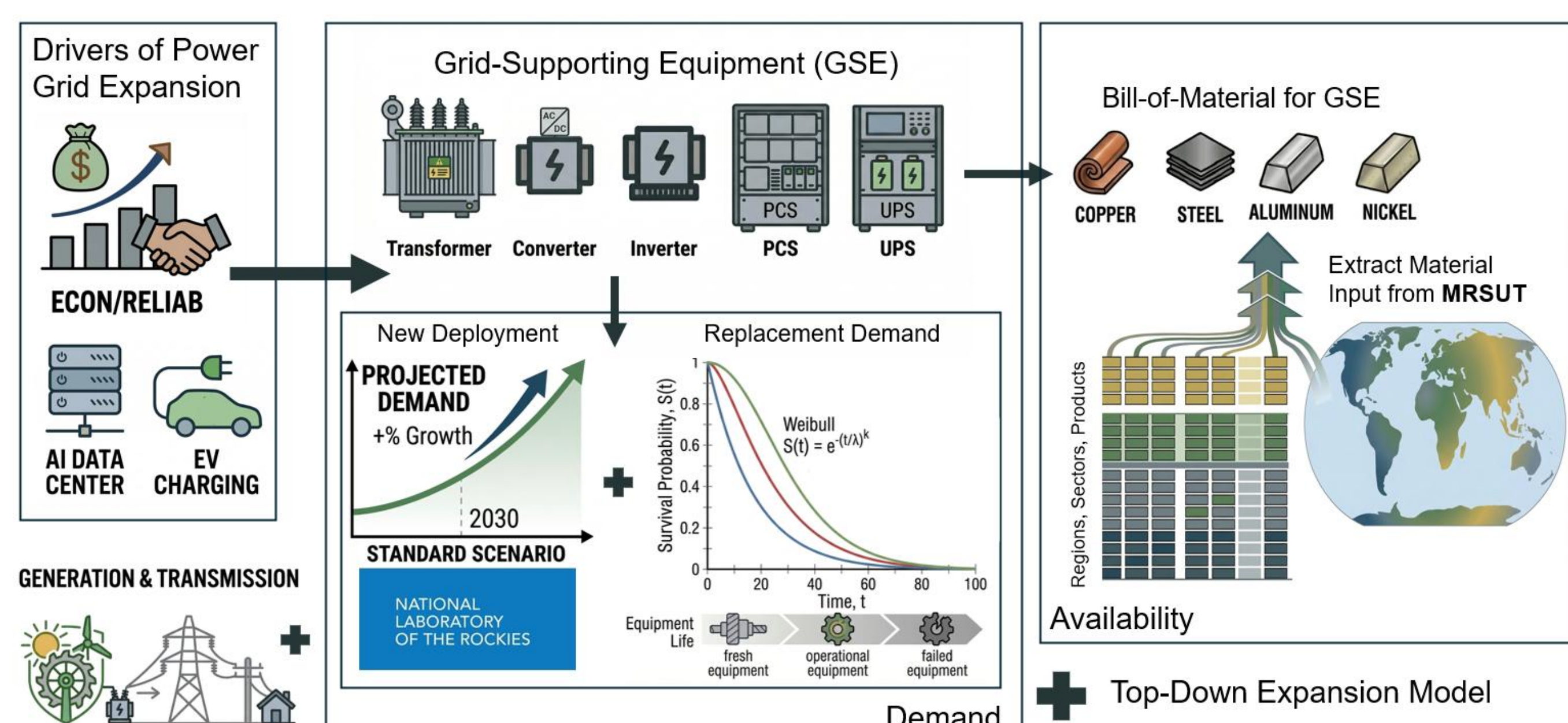
Motivation & Research Questions

Rapid load growth is increasing pressure on grid expansion. AI-oriented data centers alone could consume **9–17%** of U.S. electricity by 2030, while transformer lead times have increased to as long as **2 years**. Yet power system planning often assumes that **transformers, inverters, converters, PCS, and UPS**—collectively grid-supporting equipment (**GSE**)—can be deployed when needed.

We ask:

- How will **new deployment + replacement** shape future GSE demand?
- When supply data are limited, what **proxies** can capture GSE constraints?
- Which **materials and supply chains** could constrain GSE production?
- How do GSE constraints affect the **pace of grid expansion**?
- How sensitive are bottlenecks to load growth, **trade disruption**, and **mitigation**?

Methodology



1. Dynamic GSE Demand

Grid/load growth + equipment lifetimes → annual new + replacement GSE demand

$$Q_{e,y} = N_{e,y} + \sum_{k=1}^{y-1} Q_{e,k} \Delta F_e(y-k)$$

Total GSE requirement = Net-new deployment + Lifetime-driven replacement of prior cohorts

$$\Delta F_e(t) = \frac{F_e(t) - F_e(t-1)}{F_e(t-1)}, \quad F_e(t) = 1 - \exp\left[-\left(\frac{t}{\alpha_e}\right)^{\beta_e}\right]$$

Annual failure probability = Failure during age (t-1, t]

2. GSE → Material Requirement

Compiled equipment-level BOM data → material intensity → system-scale material demand

$$R_{m,y} = \sum_e \mu_{m,e} Q_{e,y}$$

Material requirement = Material intensity (kg/MVA) × GSE capacity required

3. Multi-Regional Supply-Chain Tracing

EXIOBASE + HS concordance + UN Comtrade → embodied material demand & regional sourcing

$$\Phi = \mathbf{U} \mathbf{g}^{-1} \mathbf{V} \mathbf{q}^{-1}$$

Product-to-product supply network = Product use per industry output × Industry supply share by product

$$\Lambda = (\mathbf{I} - \Phi)^{-1} = \mathbf{I} + \Phi + \Phi^2 + \Phi^3 + \dots$$

Total embodied requirements = Final demand + Direct inputs + Indirect upstream inputs

4. Material-Constrained GSE Deployment

Material availability → Feasible GSE deployment → Unmet demand

$$\sum_e \mu_{m,e} x_{e,y} \leq S_{m,y} \quad \forall m, y \quad x_{e,y} + u_{e,y} = Q_{e,y} \quad \forall e, y$$

Material used by deployed GSE ≤ Available material supply

Deployed + Unmet = Required GSE

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Understanding supply chain constraints for the US clean energy transition
npj Clean Energy



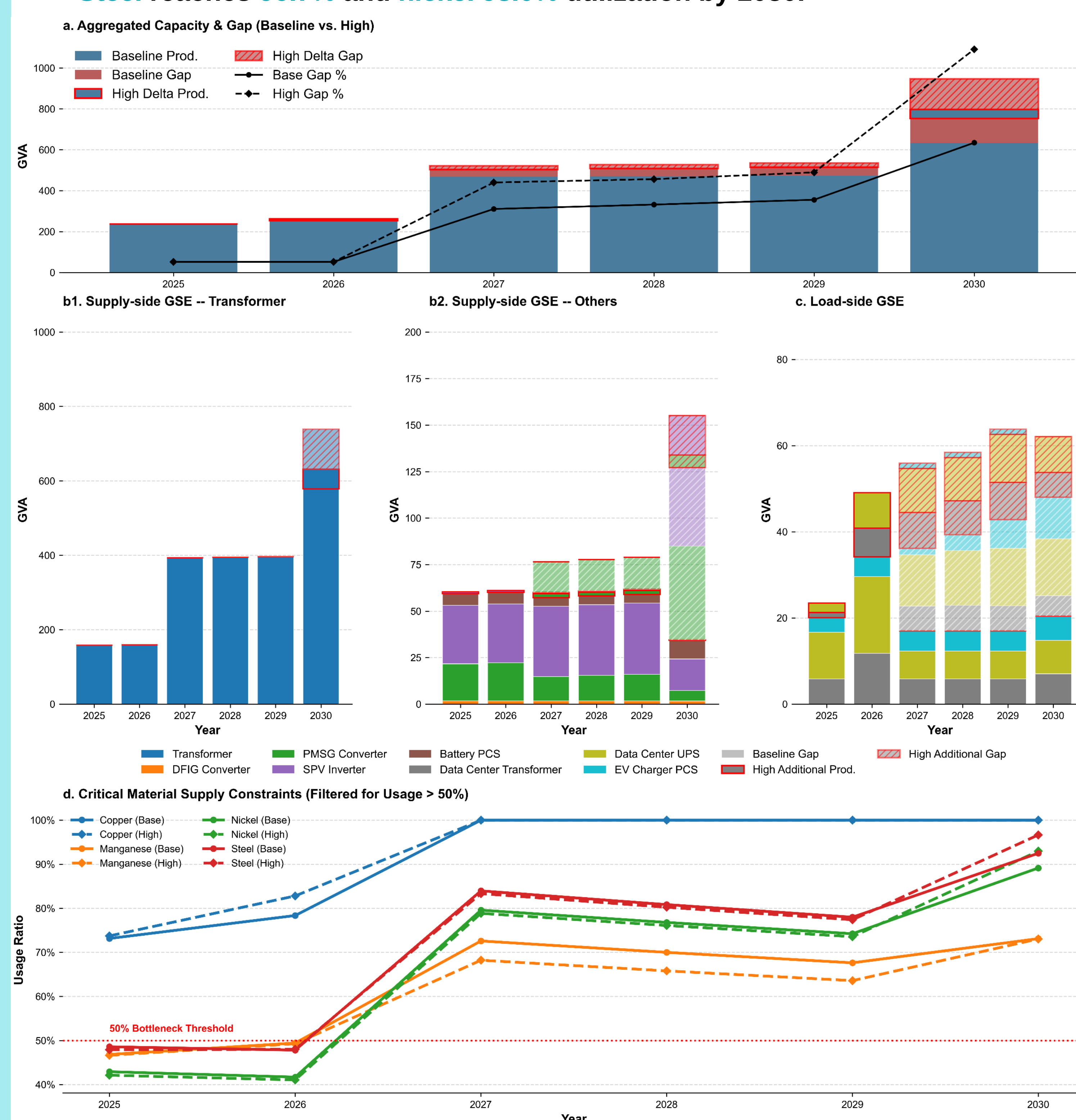
Generation Expansion Planning With Upstream Supply Chain Constraints on Materials, Manufacturing, and Deployment
IEEE TEMPR

Conclusion

- GSE supply chains can limit **grid expansion**, especially as growth increases **new and replacement demand**.
- Critical materials offer a **practical proxy** for GSE supply constraints when equipment data are limited.
- **Copper** is the key bottleneck, followed by **steel** and **nickel**.
- Resilience measures differ: **trade disruption worsens shortages**, while **DTR mainly eases transformer** constraints.
- **Planning** should explicitly account for **GSE, replacement, and material** constraints.

Results

- GSE shortages emerge **after 2026** and widen rapidly toward **2030**.
- **28.5%** of GSE demand is unmet by 2030 under high growth.
- Shortages appear first in non-transformer and load-side GSE, then extend to transformers.
- The 2030 **transformer gap** reaches **107.3 GVA (14.5%)**.
- **Copper** becomes fully **binding** from 2027 onward.
- **Steel** reaches **96.7%** and **nickel 93.0%** utilization by 2030.



- **Trade disruption worsens shortages**, increasing unmet GSE demand by **73.0 GVA to 342.6 GVA (36.2%)** in 2030.
- **Dynamic transformer rating provides partial relief**, reducing unmet demand by **75.6 GVA to 193.9 GVA (22.3%)**, but leaving most non-transformer shortages unresolved.

Metric	Trade disruption	DTR
Change in total unmet GSE demand (GVA)	+73.0	-75.6
Change in total unmet GSE demand (%)	+7.7	-6.2
Change in transformer unmet demand (GVA)	+77.4	-73.9
Change in transformer unmet demand (%)	+10.5	-9.5
Change in all other GSE unmet demand (GVA)	-4.3	-1.8
Change in all other GSE unmet demand (%)	-2.1	-0.2
Change in nickel usage ratio (%)	+7.0	0.0
Change in manganese usage ratio (%)	+8.2	0.0